MAHAVIR PATEL

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PROFILE SUMMARY

Highly motivated computer science professional with a **Master's degree in Applied Computer Science**. Proficient in **C++, Java, and Python**, with expertise in **Database Technology, Software Development, and Agile methodologies**. Strong track record of accomplishment in computer science. **Diligent, adaptable, and results-driven with Exceptional communication, problem-solving, and collaboration skills**. Seeking to contribute expertise to a dynamic organization.

EDUCATION

Master of Applied Computer Science, Concordia University 2022 - Present
Relevant Coursework: Advanced Programming Practices, Distributed Systems Design, Comparative Study of Programming Languages, Software Engineering Processes, Computer Networks and Protocols, Algorithm Design Techniques, Program and Problem Solving, Software Design Methodologies, Artificial Intelligence (*CGPA - 3.66/4.3*)

Bachelor of Engineering, Information Technology, Gujarat Technological University 2017 - 2021
Relevant Coursework: Analysis and Design of Algorithms, Data Structure, Advanced Java, Computer Networks, Operating System, Software Engineering, Computer Graphics, Image Processing, Big Data Analytics, Artificial Intelligence, Python Programming, Data Mining and Business Intelligence (*CGPA - 9.46/10*)

TECHNICAL SKILLS

Programming Languages	C , C++, J2EE (Java), JavaScript, Python, SQL, Nodejs, HTML, CSS
Frameworks	Akka, Play framework, Express.js, Java MVC, REST API, CI/CD, Object-Oriented Design, Django, Spring, Spring Boot
Methodologies	Agile Software Development, Scrum, SDLC (Software Development Life Cycle), Junit, Waterfall, Iterative methods, Unified Modeling Language (UML), Socket Programming, RESTful Services, Version Control System, Object Oriented Programming, Microservices
Databases / Tools	SQL, MySQL, SQLite, PostgreSQL, NoSQL, MongoDB, RDBMS, Firebase, JIRA, Elasticsearch, Kibana, Logstash, GitHub, Git Bash, Google Cloud, AWS, Google Colab, Postman, Docker, Linux/Unix, Visual Studio, Eclipse, IntelliJ, Linux, MS Office, Google Workspace, WAMP Server, Eclipse, Jupyter Notebook, PyCharm, VS Code, MS Word, MS Outlook, MS Excel and MS PowerPoint

WORK EXPERIENCE

Systems Engineer Aug, 2021 – Dec, 2021
Tata Consultancy Services Limited *Ahmedabad, India*

- Developed a Full stack web application for the health care sector to reduce the dispute resolution time by 60%.
- Designed backend data structures in node.js to maintain records of clients and added reactive webpages(UI) using react.js.
- Learned the agile development method and implemented it across the entire team with weekly clients meeting from the various counties on a virtual platform to implement continuous integration and manage the project with a dynamic, collaborative, progressive and highly performing team.

Intern Jul, 2020 – Sep, 2020
Tata Consultancy Services Limited *Ahmedabad, India*

- Did a remote internship under the project of scalable search engine hosted on IaaS primarily focused on the entropy of the search result.
- Learned and Used ELK stack to give the analytics of data from various companies.

- Developed a dashboard using the Kibana giving the insights of data trends with reactive graphs on the Elasticsearch cluster. Injected thousand of data from various sources with the Logstash Utilization.

PROJECTS

FreeLanceLot

- Created a reactive server-push web application using Play Framework based on Akka, scala and actor framework that examines the content of the Freelance.com website via its REST API.
- Added web Socket that dynamically updates the user interface without any user interaction via streaming the data from the website.
- Implemented session management, caching mechanism and other modules of the application using Akka framework. Used Junit unit testing and JACOCO for code coverage.

Plagiarism Detector Tool

- Implemented Cosine Similarity Index, Jaccard Index, and Word Frequency Count algorithms to calculate plagiarism or similarity rate between two input text files.
- Improved the efficiency of the algorithms by pre-processing data to remove special characters, punctuation marks, and white spaces, new line characters, and very common English literature words like a, an, the, etc.
- The plagiarism detector tool can be run using the Makefile "run" rule and can be tested on any dataset including source code files on any program.

Endangered Bird Species Detection

- Developed the AI model to categorize the endangered bird species by starting from scratch with the ResNet CNN Model using PyTorch.
- Achieved 86% accuracy compared to the ResNet pre-trained AI model's 96% accuracy.
- Completed the task using a Decision Tree with supervised and semi-supervised learning using scikit-learn, and compared the results using tSNE plots.

Online Attendance System using Face Recognition

- Developed a system that takes attendance of students/employees by using facial recognition method.
- Used machine learning algorithm, TensorFlow and OpenCV to train the model and produce the results with 80% accuracy, also incorporated database to generate various reports.
- The system was designed and developed under the Gujarat Industrial Hackathon 2018-19 and selected for the finals.

CERTIFICATIONS

- Agile for Beginners, Tata Consultancy Services Limited, Ahmedabad, India
- Python Programming, University of Michigan link
- Introduction to Data Science, University of Michigan link

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Gujarat Industrial Hackathon 2018-19: cleared the regional round and selected for the finals at PDPU University.
- Gujarat Industrial Hackathon 2019-20: cleared the regional round and selected for the finals held at CHARUSAT
- Smart India Hackathon 2020 : cleared the regional round
- TCS Codevita Season 8, 2019 – cleared first round in TCS codevita global coding competition
- Infosys HackWithInfy 2020 – cleared the first round and appeared as the top 6% candidates.
- Awarded with Academic Excellence Award for securing first rank during Bachelors.
- Achieved an award of Shri Dewang Mehta IT Award – 2019 and honored by Shreee Dewang Mehta Foundation Trust.

LEADERSHIP

- General Secretary for the term 2019-2020 at A. D. Patel Institute of Technology, Anand.
- Information Technology department representative for student central committee at A. D. Patel Institute of Technology, Anand.